

1a	Using point B: Apply $PV = nRT$ $(1.00 \times 10^5)(1.00 \times 10^{-3}) = n(8.31)(300)$ $n = 0.0401 \text{ mol}$ Using point C would have yielded $n = 0.0402 \text{ mol}$.	1 1
1b	For process AB, temperature decreases from 350 K to 300 K, thus the internal energy decreases. $\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} (0.0401)(8.31)(300 - 350)$ $= -25.0 \text{ J}$ Change in internal energy is 25.0 J.	1
1c	For process AB, the volume remains constant. So work done on gas is zero. From 1st Law of thermodynamics, the decrease in internal energy is also the amount of heat released by the gas during process AB. $\Delta U = Q + 0$	1
1d	Work done on gas from B to C, $W = p_1(V_1 - V_3) = (1.00 \times 10^5)(1.00 \times 10^{-3} - 1.17 \times 10^{-3}) = -17.0 \text{ J}$ Work done by the gas from B to C = +17.0 J Increase in internal energy from B to C, $\Delta U = +25.0 \text{ J}$ Heat supply to gas from B to C $Q = \Delta U - W = 25.0 + 17.0 = 42.0 \text{ J}$	1 1 1
1e	Heat is released. Net change in internal energy is zero for a cyclic process. i.e. $\Delta U = 0$ Work done on gas during CA is larger than work done by gas during BC. (No work done during AB). Thus W is positive. $\Delta U = Q + W$ $Q = -W$ thus as Q is negative, Heat is released by the gas during the cycle ABCA.	0 1 1

2021

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

2a

An oscillatory motion where acceleration